Cerebral oximetry is a monitor that uses near-infrared optical spectroscopy (NIRS) to noninvasively measure the regional blood hemoglobin oxygenation saturation (rSO2). It is useful in patient populations at risk for stroke such as vascular or cardiac surgery patients, patients with a history of stroke, and neurosurgical patients. The technology used in cerebral oximetry is similar to that used in pulse oximetry.

To use the monitor, a noninvasive sensor is placed over the forehead, but away from the hair and scalp. The sensor can be placed on the right, left, or both frontal-temporal regions (preferable). The sensor emits low intensity light that penetrates the skin, skull, dura, and CSF into the blood located within the cerebral cortex. It is then reflected back to the skin sensor where the rSO2 is determined based upon the light intensity changes as it passes through the brain. The light emitted is of two wave-lengths, 730 nm and 805 nm, emitted in alternating fashion through two light-emitting diodes. Cerebral oxygen saturation is related to three factors:

1. The light intensity at the source
2. The detector
3. The length of the light path through the tissue according to the Beer law (The Beer law states that there is a logarithmic dependence between the transmission of light through a substance and the product of the absorption coefficient of the substance, and the distance the light travels through the material.)

An rSO2 value of 70% would be considered normal because this encompasses both venous, arterial, and capillary blood. An index value of less than 50, or a change larger than 20%-30% would be indicative of neurological change.

Factors that can result in rSO2 desaturation include: the administration of IV indigo carmine, cardiac arrest, hypothermia, blood steal syndromes, hypoxia, ischemia/embolic strokes, Trendelenburg position (leads to cerebral venous congestion), unilateral or bilateral cerebral atherosclerosis ,and the interference of hair, scalp, or hair follicles. A left-shifted curve can create false positives, and some patients have normal neurological function, yet have low baseline values.

Sandberg W, Urman RD, and Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier; 2011: 142-143.

Copyright 2015 Prodigy Anesthesia. It is unlawful to copy, distribute, and/or sell this material.